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Psych 124J

Introduction

About 3% of children are diagnosed with autism, 11.4% are diagnosed with ADHD, and 15% of people have dyslexia. Many find these diagnoses extremely inhibiting, especially in academic learning settings. However, it is evident that these populations can learn. For example, many students with ADHD can still attend college and be successful, and some autistic students are still capable of learning how to read or basic math. One important aspect of learning is category learning, as it helps reduce cognitive load by allowing us to generalize features instead of brute memorization. It is also integral in academic learning, as all subjects have rule-based patterns. How does category learning differ from each population? This paper analyzes three research papers regarding category learning for autistic, ADHD, and dyslexic populations.

1. Category Learning in Autism

Overview

This study examines how feedback intensity and information presentation played a role in autistic children's ability to learn in probabilistic category learning tasks.

Methods

Subjects had to categorize which artificial customer wanted which type of ice cream or juice. Researchers created 14 different, artificial characters using either Mr. Potato Head, who wanted chocolate or vanilla ice cream, or Ms. Pumpkin who wanted grape or apple juice. All characters had basic features (arms, legs, nose, mouth), and the 14 characters were created by the varying presence/absence of certain features pertaining to the character. Each feature was associated with an outcome following a fixed probability; for example, moustache and bowtie were highly predictive of outcome, and subjects could accurately predict which ice cream the customer wanted based on looking at these two features; however, only considering hat or glasses would decrease the probability of accurate prediction. Each child was assigned to either feedback or information presentation, and completed both conditions within each category.

The low-intensity feedback consisted of simple, informative information, whereas the high-intensity feedback consisted of informative information, but with extra, nonsocial information as well. The isolated condition in the information presentation condition consisted of one item being shown at a time, in a sequence whereas the simultaneous sequence consisted of 20 at a time, in an array.

Figure 1
Feedback Condition Stimuli

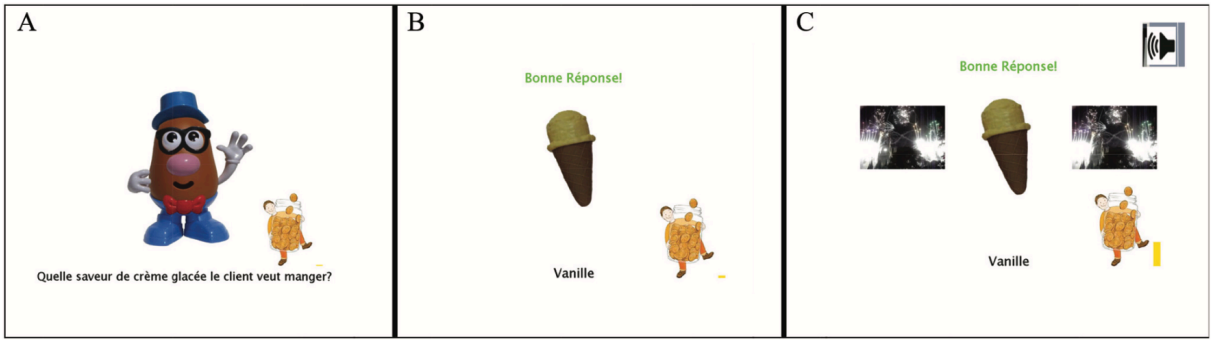


Figure 1

Figure 2
Information Condition Stimuli

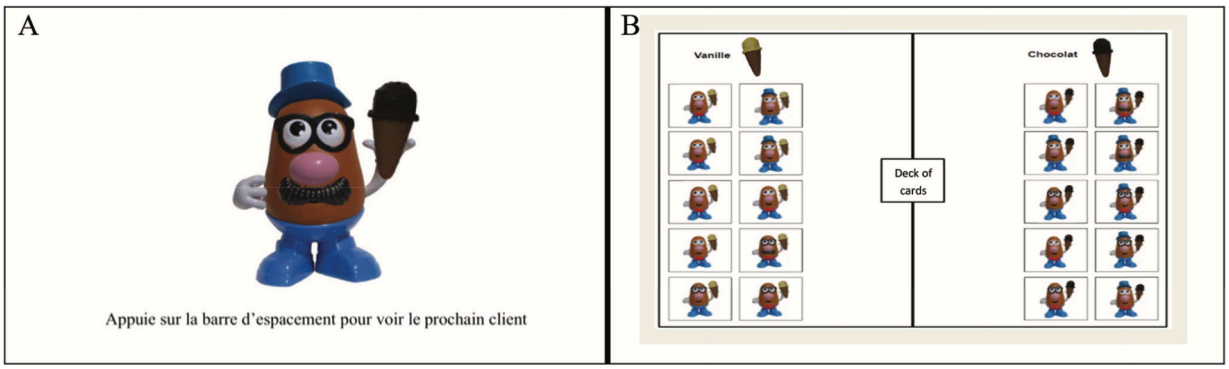


Figure 2

There were two test phases. Test 1 used the same stimuli as the learning phase (Mr. Potato Head and Ms. Pumpkin), and Test 2 used similar but equivalent stimuli (same probabilities associated with features present or absent, but different colored or sized features). This attempted to determine whether participants could generalize learning. Testing generalization in this task is particularly important, as it ensures that the learning done by both groups is not a memory task.

Results

Autistic children performed the same in both feedback situations, indicating that feedback for autistic students is not as important as initially thought by researchers. Typically developing children performed significantly better than autistic children when they received high feedback, which shows that typically developing children benefit from high-feedback. In fact, autistic children's accuracy slightly decreased with high feedback.

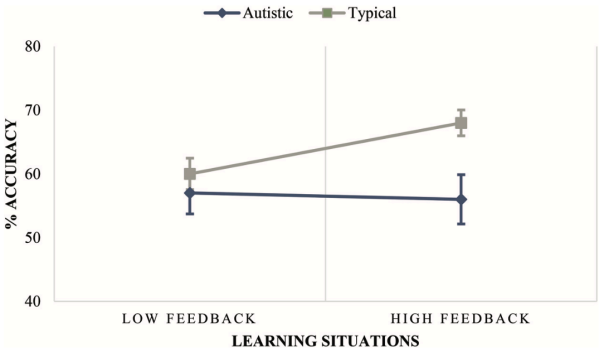
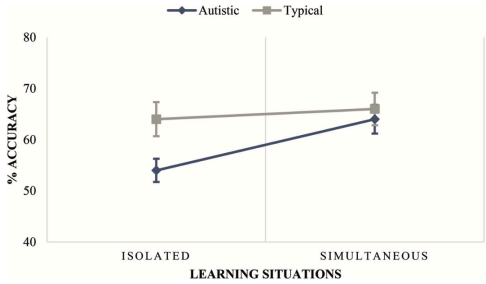


Figure 5

However, it is unclear what exactly with high feedback helps typically developing children. Do they remember how the extra visual and auditory cues relate to the correct answers better, or is there a neurological pathway stimulated, helping them remember and apply the pattern better? Is this not present or not relevant for autistic children? Or, maybe the autistic children struggle to associate the given positive cues as positive feedback. I propose that "high feedback" might be different for each autistic student, especially when thinking of how parents try to encourage their autistic child. For example, an autistic child might only associate receiving extra play time as a reward for doing something correctly, so positive audio or visual stimuli might be meaningless. However, if given a stimulus of their choice rewarding them for getting something right, the results might have been different.

Both groups performed better when they had information presented to them simultaneously, with autistic children significantly improving accuracy when presented information simultaneously. The graph below shows how both groups are almost at the same level of accuracy when presented with simultaneous stimuli.



This could indicate that isolated information is not helpful at all in this task at all. For example, giving children simultaneous information with regards to math questions might have a different effect. The way the simultaneous information was presented is very specific to this type of study only, and cannot be generalized to other types of learning. Therefore, the notion that simultaneously presented information is always better for autistic students should be cautioned against.

One of the main conclusions of the paper is that contrary to well-known research, autistic students do not gain much from extensive feedback; rather, they "benefited from a feedback-free situation where they have access to large quantities of simultaneously presented complex information." However, the type of information presented (pictures of a familiar character) in this manner might be the reason both autistic students and typically developing students performed better. Presenting any information simultaneously might not be the answer to helping

students with autism. Additionally, maybe feedback has to be relevant to the child to make a difference.

2. Rule-based and Information-Integration perceptual category learning in children with Attention Deficit Hyperactivity Disorder

Overview

This study aims to understand whether children with ADHD have a deficit for rule-based learning and information integration learning, as previous literature suggests that ADHD is linked to executive function (attention and inhibition). Researchers also want to understand if these deficits are because of a general inability to process multiple dimensions or if they are specific to associative learning systems (implicit learning).

Methods

This research paper uses two different studies to explore the initial questions. Study 1 tested rule-based versus Information-Integration learning. All subjects completed a RB task that could be solved by sorting along spatial frequency and an II task in which both spatial frequency and orientation were required for optimal performance. The tasks were spaced one week apart and were counterbalanced. The task itself had MATLAB designed Gabor Patches, which varied in how slanted the stripes were (orientation) or how fine/coarse the stripes were (spatial frequency). Subjects had to learn to recognize whether the patches were in Category A and B; feedback was simply "correct" or "wrong." The RB task had only one dimension altered, and the child was supposed to infer an explicit rule such as "if the patch has coarse lines, its Category A." The II task required subjects to categorize both frequency and orientation, and there is no simple rule, testing implicit associative learning. The notion that there were no learning or testing phases is necessary because it allows researchers to determine how people form categories, as well as if and when they switch strategies. Additionally, there were 5 blocks with 80 trials each, 40 of which the stimuli belonged to Category A and 40 belonged to Category B. This study design is very thorough, as conditions were counterbalanced, and tested in a manner that is simple yet effective.

Results

With regards to the results, the paper states "Children with ADHD were more hyperactive, impulsive, and inattentive than controls" pg 597. This statement does not seem supported by the data, as the data shows the proportion of whether or not children with ADHD were able to learn and recognize RB or II patterns; it does not seem accurate to make generalized statements about impulsivity or hyperactivity, which are more behavioral or neurological observations. In the RB task, both groups learned the patterns, with the control group outperforming the ADHD students. One interesting and initially confusing aspect about the data is the use of specific models to analyze the data. Researchers used a GLC model to determine if students were using II strategies in the RB task, even though II is not necessary for the RB task. They also used a random model to determine if subjects were using random strategies. When analyzing whether the two groups

used relevant vs irrelevant strategies, it seems that the students with ADHD gradually increased their ability to use relevant strategies and decreased the use of irrelevant strategies, whereas the control group had a spike in the use of irrelevant strategies in Block 4. By Block 5, about half of the control group was using the relevant strategies, greater than the ADHD group. This is odd, as the steep change for the control group is large enough.

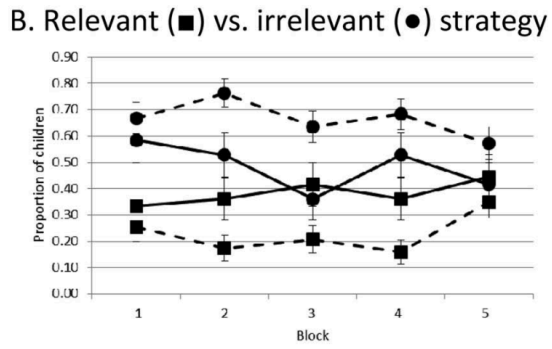


Figure 2B

Overall, it appears that children with ADHD were able to identify relevant strategies, and did not use random strategies in the RB task. With the II strategy, both groups improved their accuracy from Block 1, with the control group showing more consistent accuracy whereas the ADHD group had a slight dip in accuracy at Block 5. However, children with ADHD used random strategies more than the control group for the II task. In Block 4 of Figure 1H, it is interesting how around Block 4, both groups are almost at the same level, however the two distinctly diverge by Block 5. For two groups that are generally seen as extremely different, this similar proportion of children using random strategies is intriguing.

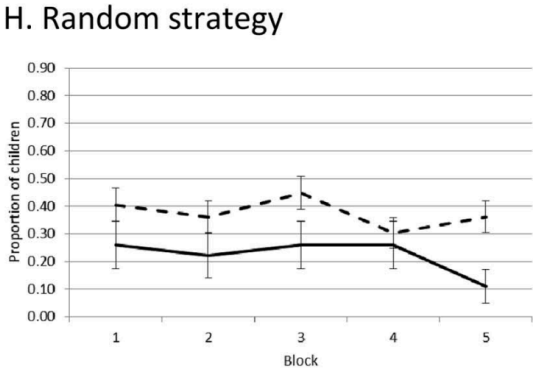


Figure 1H

It is extremely admirable that Study 1 used various models to understand how and why the groups opted for a specific strategy. Simply knowing that the controls outperformed the ADHD group is not as informative as understanding what strategies the majority of children opted for.

Study 2 attempts to separate out implicit learning deficits (II) and general inability to use multi-dimensional rules. The conjunctive rule task is explicit, as it requires children to consider two stimulus dimensions, but there is an explicit rule: if both spatial frequency and orientation

are above/below a certain threshold, then they are Category A, otherwise they are Category B. The stimuli were the same as Study 1 (spatial frequency and orientation), however the variances of the two dimensions were equal; essentially, both stimuli were equally as important, whereas in Study 1, one was more important than the other. This study used 7 blocks with 54 trials each, and had the same visual "correct" and "wrong" feedback. A completely new set of participants were recruited. Once again, the paper states "children with ADHD were more hyperactive, impulsive, and inattentive than controls," which is a strong statement. The data shows that the ADHD group performed significantly worse than the control group in terms of accuracy, however for the optimal CJ model, few control group participants were able to use the optimal rule, and even fewer in the ADHD group used it. This might suggest that the CJ task is difficult for most students, regardless of ADHD. Figure 1B is extremely intriguing, as more students with ADHD used the optimal strategy compared to the control, and the control group's use of the optimal strategy is inconsistent. However, when examining the accuracy graph, it is evident that the control group is still more accurate than the ADHD group.

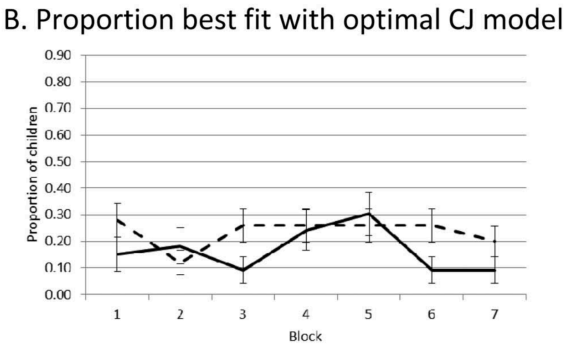


Figure 3B

Controls showed that they more consistently use unidimensional strategies, which is still suboptimal, yet at least somewhat RB. This might be the reason for the control group having higher overall accuracy. On the other hand, the ADHD group was less consistent and less likely to use these strategies. With regards to the random model in the CJ task, the ADHD group decreased in using random strategies, however the proportion of children using the random strategy increased by Block 7.

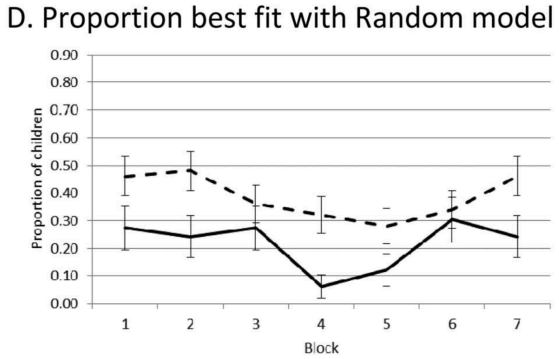


Figure 2D

The proportion of students in the control group who used random strategies was less than the ADHD group, however the pattern is generally inconsistent. Similar to Study 1, both groups had

almost an equal proportion of children using the random strategies at Block 6, however it diverges by Block 7. This might suggest that maybe children try different strategies with each Block. However, it is interesting to see these patterns in the data, because why are children using random strategies if they have already been given feedback about the categories? Because the CJ task is explicit, this shows that the children with ADHD struggle with multidimensional stimuli in addition to implicit learning.

Conclusion

The paper concludes that overall, children with ADHD have a deficit in associative II learning and explicit RB learning. However, it is worth noting that some children with ADHD still have the ability to learn patterns and use optimal strategies.

3. Reduced Categorical Learning of Faces in Dyslexia

Overview

This study revolves around dyslexic individuals, and stems from the discrepancies in previous studies regarding dyslexia and facial recognition. It tests dyslexics ability to categorize faces and implicit learning.

Method

Participants were asked to categorize female faces as “Daphna” or “Liat,” and were shown faces of the women. 0% was Daphna, and 100% was Liat; the faces were morphed by 30%, 40%, 50%, 60%, and 70%. In the introduction or learning phase, only the two endpoint faces were shown, and if the accuracy was below 85%, researchers made participants try again. Each face was presented 10 times in a random order, was presented for 1.5 seconds, and participants were given feedback. In the experimental phase, only the morphed faces were shown, each appeared 139-148 times, which appears random. The ITI or inter-trial interval was also manipulated- 2, 3.5, or 5 seconds, to determine if the participants benefit from longer ITIs, as longer ITIs reduce attentional load. In this phase, no feedback was given. This is quite intriguing, as previous papers analyzed gave participants feedback to help them learn, but this paper forces all participants to remember what they learned in the introduction phase to aid them in all of the experimental trials. This draws in implications of dyslexia and memory. Had participants been given constant feedback with each of the trials, they may have had to rely on less memory, however there could have been other unknown implications.

Results

To calculate how well participants were able to discriminate between the two faces, researchers used d’=hit rate- false alarm rate. Both groups showed improvement between sessions, with the typically developing group outperforming the dyslexic group. If a participant simply forgets,

then all of their trial data might as well be guessed data.

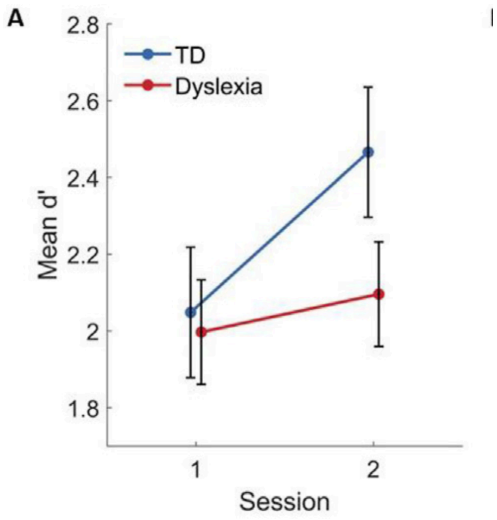


Figure 3A

The 50% condition was used to determine bias, as there is no correct answer for this one. Results showed that both groups were not biased towards one name or the other. In terms of the reaction time, TD participants had a lower RT than the dyslexic group, however both groups showed a decrease in RT between sessions, suggesting that there is some learning done in both groups. Additionally, researchers employed a Drift-Diffusion-Model, which is a mathematical model used to show decision making, and explains both accuracy and reaction time. Essentially, the higher the drift rate, the higher the speed and accuracy of decisions. Data shows that TD's increase in drift rate between sessions was significant, however for the dyslexia group, the increase was present but not significant.

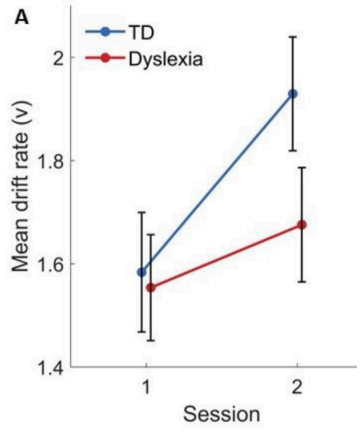


Figure 6A

This suggests that although both groups start at the same spot, learning is impaired for the dyslexic group. Another factor in the DDM is the decision boundaries, which shows how much evidence one requires before making a decision. A higher boundary means one is more cautious, whereas a lower boundary indicates a willingness to guess faster. No significant group differences were found in boundary settings. Lastly, the non-decision time in the DDM

represents time spent on processes not involving decision making, such as visual encoding or motor execution. The dyslexic group had longer non-decision time, which is partially the reason for their longer reaction time. In terms of different ITI's, both groups improved their ability to categorize with longer ITIs, and there was no significant difference between the two groups.

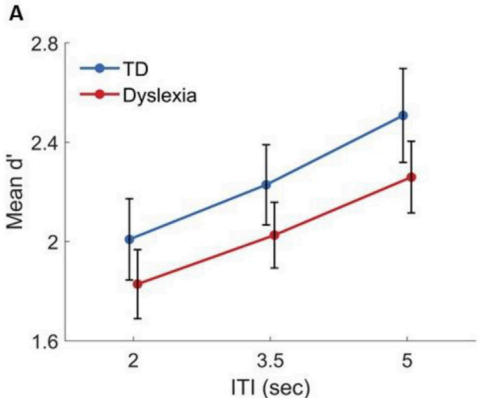


Figure 7A

Taken together, these data suggest that the slower category learning of new faces among dyslexics shows a core learning deficit. However, dyslexics still learned to recognize faces; they just did so at a slower rate than typically developing participants. Dyslexic people do recognize people in general, but they need more, repeated exposure to achieve the same as typically developing individuals. Additionally, participants were not given trial-by-trial feedback, which could have large impacts on the results of the experiment. With more feedback, maybe dyslexics learn better, however not giving them feedback also exposes the natural impairment with learning.

Conclusions

Although this paper examined different neurodevelopmental disabilities, there are some striking relationships between them.

1. In all of the papers, all participants learned, just at different rates.
2. Feedback is not necessarily useful- for autistic and ADHD participants, feedback did not seem to have a strong impact on learning ability. For the dyslexic research paper, feedback was only present in the learning phase, and it appears that participants still learned without constant feedback in the experimental phase.
3. ADHD and dyslexics show an implicit learning deficit.
4. Multidimensional category learning is less in ADHD and dyslexics

All of the participants in the studies learned. In the first paper, autistic students were almost at the same level of accuracy as the control group when presented with simultaneous information. This shows how for autistic students, there can be an optimal strategy for helping students learn in the most efficient way. In the second paper, students with ADHD learned, however significantly less than the control group. In Study 1's rule-based experiment, the ADHD group rarely used random strategies, and the proportion of children using the relevant strategies by the last block increased from the first. This reveals how they are capable of rule-based explicit learning. Lastly, dyslexic students were able to discriminate between faces

at a higher rate than from when they started, and showed improvement when given more time between intervals. Although lower than the control group, it still shows their ability to learn categories.

Traditionally, many programs for neurodevelopmental disabilities place a strong emphasis on feedback, as previous literature suggests that more feedback equates to better performance. However, the first study clearly showed that auditory and visual feedback does not really make a difference for autistic students. In the second paper, feedback was simply "right" and "wrong," similar to the low-feedback condition in the first paper. This feedback did not seem to play a strong role in ADHD participants' ability to learn. In the last paper, feedback was only given in the introduction phase. It is difficult to speculate how different the results would have been if feedback was given throughout the experimental phase. However, it is worth noting that even without feedback, dyslexics still learned, albeit at a lower rate and accuracy than the control group.

With regards to implicit learning, the first paper showed that implicit learning is somewhat intact with simultaneous input, however for ADHD individuals and dyslexic individuals, there is a deficit in implicit learning, as the ADHD group struggled in the II taks, and the dyslexic individuals had a slower d' and drift rate when compared to controls.

All of these experiments also examined how individuals can integrate multidimensional stimuli. The first paper revealed that autistic students performed better when they integrated multiple features into one stimuli, however the second paper showed that when students have to integrate multidimensional information (CJ Task), they struggle. In the last paper, faces have multiple dimensions that need to be integrated to distinguish between two different faces, and there was slower learning seen. However with the last one, there was no unidimensional condition, so an extremely conclusive statement cannot be made. Regardless, it is evident that ADHD and dyslexic populations struggle to integrate multiple dimensions to learn categories.

Future Research

One idea for future research is to experiment with differential feedback modalities for these three populations. One popular notion is that these groups are often discouraged by learning because of social pressures, however these research papers do not consider feedback as an important aspect of learning. Additionally, the strength of feedback can vary, from no feedback at all, to auditory and visual feedback ("Good job!" and confetti on the screen, with a "yay" auditory feature), to verbal feedback from the researchers if they are past a certain threshold for accuracy ("You are doing such a great job! Keep going" or "You are performing at a suboptimal level, do better"). By analyzing these three intensities of feedback, researchers can determine how important feedback really is for these populations. The experiment can have 30 autistic, 30 ADHD, and 30 dyslexic individuals, with 90 matched peers. Each condition will have 20 individuals, 10 of each developmental disability, and 10 typically developing individuals. The task can be spatial frequency and orientation, similar to the second study. There should be a learning and testing phase with the testing phase being different enough from the learning phase to assess generalization. I hypothesize that autistic students might benefit from extremely personable feedback. For example, if the student likes cars, giving them a car after each trial they do well on and

taking one away if they do suboptimal might have an impact on their learning. One caveat is that if the kids are too young or emotionally immature they might have a meltdown if someone takes away their toy. However, I acknowledge that highly personable feedback might be hard to implement for each child, in which case I hypothesize that generalized feedback for autistic students might not play a strong role in their ability to learn. I hypothesize that students with ADHD might benefit from frequent verbal praise because it might reinforce their attention, possibly helping them learn better. I believe that dyslexic individuals might benefit from strong praise, however they might rely more on feedback and repetition. Typically developing individuals will probably benefit from medium or high feedback.

Another idea for future research is integrating multiple dimensions for all three populations. The same 30 autistic, 30 ADHD, and 30 dyslexic individuals, with 90 matched peers can be assigned to unidimensional and multidimensional conditions. The stimuli can be distinguishing faces. In the unidimensional task, only a feature like nose size can matter, whereas in the multidimensional nose size and eye size can matter. I hypothesize that autistic students will do well with both, but better at the multidimensional task, ADHD and dyslexic individuals will do better with the unidimensional feature.

Together, these proposed experiments would help us better understand how feedback type and dimensional complexity interact to influence category learning across these neurodevelopmental conditions. These findings could potentially help educators and parents identify better ways to help these populations learn in the most efficient way possible.